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|-----------|--------------------------------------------------------------------------------------------------------------------|-----------|
|           |                                                                                                                    |           |
| 3(a)      | (thermal) energy per unit mass                                                                                     | <b>B1</b> |
|           | energy to change state between liquid and gas at constant temperature                                              | <b>B1</b> |
| 3(b)(i)   | $q = mL = 0.37 \times 2.3 \times 10^6$<br>$= 8.5 \times 10^5 \text{ J}$                                            | <b>A1</b> |
| 3(b)(ii)  | $pV = nRT$ and $T = 373 \text{ K}$                                                                                 | <b>C1</b> |
|           | $n = 370 / 18$                                                                                                     | <b>C1</b> |
|           | $V = [(370 / 18) \times 8.31 \times 373] / (1.0 \times 10^5) = 0.64 \text{ m}^3$                                   | <b>A1</b> |
| 3(b)(iii) | $w = p\Delta V$                                                                                                    | <b>C1</b> |
|           | $= 1.0 \times 10^5 \times 0.64$                                                                                    | <b>A1</b> |
|           | $= 6.4 \times 10^4 \text{ J}$                                                                                      |           |
| 3(b)(iv)  | (water does work against atmosphere so) work done on water is negative                                             | <b>B1</b> |
|           | increase in internal energy $= (8.5 - 0.64) \times 10^5 = 7.9 \times 10^5 \text{ J}$                               | <b>A1</b> |
| 3(c)      | valid reasoning of how work done by water is affected                                                              | <b>M1</b> |
|           | correct use of first law to draw conclusion about effect on specific latent heat that is consistent with work done | <b>A1</b> |